ArPay

Eco-Incentive Settlement Protocol

Green Reward Distribution Infrastructure

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Abstract

We propose a trustless settlement protocol that enables the conversion of on-chain USDC holdings into local fiat currency disbursements, without requiring local community hubs or environmental program receivers to hold, manage, or interact with any cryptocurrency asset. The system is designed to serve as Green Reward Distribution Infrastructure: a protocol layer that allows DePIN networks, sustainability sponsors, and environmental grant programs to disburse incentive rewards in Indonesian Rupiah directly to grassroots community receivers — such as registered recycling cooperatives, community composting hubs, or air-quality sensor node operators — who operate entirely within the domestic banking system.

The system leverages Solana's sub-second block finality as a cryptographic commitment layer, a Program Derived Address (PDA) for atomic fund escrow, and an off-chain oracle bridge that listens for on-chain events and triggers licensed fiat disbursements via the domestic payment network. The community receiver exclusively receives legal tender; the eco-sponsor exclusively signs a standard blockchain transaction from a self-custody wallet. Neither party requires custodial intermediaries. The total incentive disbursement cycle — from transaction signature to community hub bank credit — is demonstrated to complete within five seconds under nominal network conditions.

1. Introduction

Distributing green incentive grants to local environmental initiatives remains hampered by a fundamental structural mismatch. Local communities — recycling cooperatives, DePIN sensor node operators, urban greening collectives, and waste-reduction programs — operate entirely within a banking system that recognizes only legal tender. Sustainability sponsors and DePIN networks, meanwhile, hold value denominated in programmable, censorship-resistant tokens optimized for borderless, low-cost transfer. No direct settlement mechanism currently exists that satisfies both parties' native operational requirements without demanding that one adopt the infrastructure of the other.

Existing solutions universally fail at the community boundary. Centralized exchanges require the eco-sponsor to surrender custody, complete identity verification, and accept settlement delays measured in hours or days. Eco-reward distribution pipelines that require community hubs to interact with on-chain wallets impose regulatory exposure and technical complexity that grassroots environmental organizations are neither equipped nor willing to bear. Point-of-sale integration plugins require community receivers to understand and accept novel financial instruments — a barrier that effectively excludes the very populations most aligned with green initiatives. The result is near-zero adoption of crypto-denominated sustainability grants at the community level.

ArPay resolves this asymmetry through a protocol-level separation of concerns. The eco-sponsor-facing settlement layer operates entirely on Solana's high-throughput

blockchain. The community-facing disbursement layer operates entirely within the existing domestic banking system. A cryptographically-verified bridge — implemented as a Python daemon monitoring RPC events — connects the two layers without requiring trust in any single party.

The protocol is designed specifically for the Indonesian QRIS ecosystem, where a standardized QR code payload encodes the community hub's national identifier and the incentive amount, enabling automated extraction of all parameters required to route a fiat disbursement to the correct community bank account. The mechanism generalizes to any market with a standardized QR payment scheme and a licensed Payment Gateway API.

2. Background and Prior Work

2.1 The QRIS Standard

Quick Response Code Indonesian Standard (QRIS) is a unified QR payment standard mandated by Bank Indonesia (BI). The QR payload conforms to the EMVCo Merchant Presented QR specification and encodes, among other fields, the National Merchant ID (NMID) and the transaction amount. QRIS is accepted by over 30 million registered terminals in Indonesia as of 2024, making it the dominant point-of-sale interface for domestic financial flows — and by extension, the most practical disbursement identifier for community-level green reward distribution.

2.2 Solana as a Settlement Layer

Solana is a Proof-of-Stake layer-one blockchain employing a Proof-of-History (PoH) mechanism to achieve globally consistent block ordering without communication overhead. The network achieves practical finality within 400–800 milliseconds at a theoretical throughput exceeding 65,000 transactions per second. These properties make Solana the only public blockchain presently capable of acting as a synchronous commitment layer for real-time green grant disbursement flows that require sub-second confirmation — enabling DePIN networks and sustainability sponsors to disburse incentive rewards with the same immediacy as a conventional digital payment.

2.3 USDC and SPL Tokens

USD Coin (USDC), issued by Circle, is a fully-reserved, regulated stablecoin with 1:1 dollar parity. Its Solana implementation conforms to the SPL Token standard, enabling native integration with the Solana Program Library's associated token account model. USDC represents the ideal medium for cross-border sustainability grant disbursement: it is liquid, price-stable, and transferable at blockchain speed with negligible transaction cost — making it suitable for micro-grant flows to individual community environmental programs.

2.4 Program Derived Addresses

A Program Derived Address (PDA) is a deterministic account address derived from a combination of a Solana program's public key and one or more seeds. PDAs have no corresponding private key and can only be modified by the program from which they are

derived. This property makes them suitable as trustless escrow accounts: no external party, including the protocol operator, can unilaterally withdraw funds. The PDA serves as the cryptographic foundation of ArPay's atomic incentive disbursement guarantee.

3. Protocol Design

3.1 System Model

The ArPay protocol involves five logical participants: the Eco-Sponsor (a sustainability-aligned digital asset holder using a self-custody wallet, or a DePIN network's automated reward system); the ArPay On-Chain Program (a Solana program deployed at a fixed program ID); the Oracle Bridge (a permissioned off-chain daemon operated by the ArPay entity); the Payment Gateway (a licensed payment processor); and the Community Hub (a QRIS-registered environmental organization or community receiver with a domestic bank account). The trust assumptions are as follows:

- The Eco-Sponsor trusts the Solana network for transaction finality and the ArPay program for correct escrow behavior, both of which are publicly verifiable on-chain.
- The Community Hub trusts the Payment Gateway for fiat disbursement, which is an existing, regulated relationship predating ArPay. The community hub has no on-chain presence whatsoever.
- The ArPay Oracle Bridge is trusted to relay on-chain events to the Payment Gateway honestly. This is the sole centralized trust assumption in the system, mitigated by on-chain auditability of all fund flows.

3.2 Tri-Layer Architecture

The protocol operates across three distinct execution environments, each with defined interfaces to adjacent layers:

```
+---------------------------+
| CLIENT LAYER (L1)         |
| Next.js PWA               |
| Solana Pay URI            |
+-------------+-------------+
              | signs & submits USDC incentive tx
              v
+---------------------------+
| ON-CHAIN LAYER (L2)       |
| ArPay Solana Program      |
| Anchor / PDA Escrow       |
+-------------+-------------+
              | emits SettlementRequested event
              v
+---------------------------+
| ORACLE & BRIDGE (L3)      |
```

```
| Python Daemon / WSS       |
| Payment Gateway API       |
+-------------+-------------+
              | POST /disbursements (IDR)
              v
+---------------------------+
| COMMUNITY HUB BANK ACCT   |
| BI-FAST Green Disbursement|
+---------------------------+
```

Fig. 1. ArPay tri-layer eco-incentive disbursement architecture.

3.3 The On-Chain Program

The ArPay Solana program, written in Rust using the Anchor framework, exposes two primary instructions:

`initiate_settlement(community_id, usdc_amount, idr_amount)`

Invoked by the Eco-Sponsor. Transfers usdc_amount from the Eco-Sponsor's associated token account into a PDA derived from the tuple (program_id, sponsor_pubkey, community_id, nonce). Upon successful transfer, emits the following event:

```
SettlementRequested {
    community_id: String,  // NMID from QRIS payload
    idr_amount:   u64,     // IDR incentive amount in minor units
    usdc_amount:  u64,     // USDC in base units (6 decimals)
    payer:        Pubkey,  // Eco-Sponsor public key (for refund)
    nonce:        u64,     // Monotonic nonce for PDA uniqueness
    timestamp:    i64,     // Unix timestamp at instruction execution
}
```

`release_escrow(settlement_id, recipient)`

Callable only by the ArPay program's designated authority key, or by a time-locked refund path if no release has been issued within the timeout window (T = 120 seconds). This instruction transfers the escrowed USDC either to the protocol's treasury (upon confirmed fiat disbursement to the community hub) or back to the eco-sponsor's token account (upon disbursement failure).

3.4 Exchange Rate Oracle

The conversion rate between USDC and IDR is sourced at the moment the Eco-Sponsor initiates the incentive grant. The primary source is the Pyth Network's USDC/USD price feed, a first-party oracle operating natively on Solana with sub-second updates. The IDR/USD rate is obtained from a secondary off-chain API with signed responses. The composite rate is computed client-side and encoded in the transaction instruction data. The smart contract validates that the on-chain Pyth price at execution time is within a configurable slippage tolerance (default: 0.5%) of the rate encoded by the client.

This mechanism prevents a class of race-condition attacks where a significant price movement between rate display and transaction confirmation could cause material loss to either the eco-sponsor or the community receiver.

4. Incentive Disbursement Lifecycle

A complete ArPay green reward disbursement proceeds as follows. Timestamps T_0 through T_5 denote the sequential state transitions:

Step	Actor	Action
T0	Eco-Sponsor	Scans QRIS code at community hub. PWA decodes NMID and IDR incentive amount. Fetches USDC/IDR rate (Pyth + off-chain composite). Constructs Solana Pay URI encoding the grant instruction.
T1	Eco-Sponsor	Reviews rate. Signs and submits transaction via self-custody wallet.
T2	Solana	Block produced. Transaction executed. USDC transferred to PDA. SettlementRequested event emitted. Block status: Confirmed (~400ms) / Finalized (~1.2s).
T3	Oracle	Python daemon detects event via RPC WebSocket (WSS). Verifies block status ≥ Confirmed. Extracts community_id and idr_amount.
T4	Bridge	POST /v2/disbursements to Payment Gateway (Xendit). Payload: { bank_code, account_number, amount_idr }.
T5	Gateway	Disburses IDR incentive via BI-FAST. Community hub bank account receives credit.

Fig. 2. ArPay incentive disbursement lifecycle. Total wall-clock time T0→T5 ≈ 3–6 seconds.

5. Security Analysis

5.1 Fund Safety

Eco-Sponsor funds are at risk only during the escrow window between T2 and T5. During this window, funds reside in a PDA controlled exclusively by the ArPay program. No human key can unilaterally withdraw them. The two exit paths from the PDA are: (a) programmatic release to the protocol's treasury upon verified fiat disbursement to the community hub, or (b) automatic refund to the eco-sponsor upon timeout expiry. The program's release instruction is restricted to the ArPay authority key; the refund path is permissionless after the timeout and requires no action from the protocol operator.

5.2 Double-Spend Prevention

Solana's deterministic transaction ordering and the PDA's unique derivation from the (sponsor_pubkey, community_id, nonce) tuple prevent double-spend at the protocol level.

Each nonce is consumed atomically upon PDA creation. A replay of an identical transaction with the same nonce will fail at the program layer with an AccountAlreadyInitialized error.

5.3 Oracle Manipulation

The slippage tolerance check embedded in the on-chain program constrains the impact of oracle manipulation. An attacker who successfully manipulates the Pyth price feed beyond the 0.5% tolerance window will cause the transaction to fail at execution, not silently settle at a disadvantageous rate. The composite rate model (on-chain Pyth + off-chain signed API) requires simultaneous compromise of two independent data sources to produce a silently accepted bad rate.

5.4 Bridge Failure Modes

The Oracle Bridge is the only centralized component. Its failure modes are:

- **The daemon implements exponential backoff reconnection. Unprocessed events are recovered via slot-range RPC queries upon reconnection.**Network partition (RPC disconnect):
- **Up to five retry attempts are made with exponential backoff over a 60-second window. If all retries are exhausted, the daemon submits a signed refund instruction to the Solana program, releasing escrowed USDC to the eco-sponsor.**Payment Gateway API failure:
- **All settlement state is reconstructed from on-chain data upon restart by querying all unfinalized PDA accounts associated with the program ID.**Daemon crash:

5.5 Regulatory Isolation of Community Receivers

This is a defining design property of ArPay. The protocol’s architecture deliberately partitions all cryptocurrency exposure to the eco-sponsor side of the transaction. Community hubs — recycling cooperatives, DePIN node operators, composting collectives, or any other environmental receiver — receive only Indonesian Rupiah (IDR) via a regulated payment processor and have zero on-chain presence.

This structural isolation has critical regulatory implications under Indonesian law:

- Community receivers are not classified as Virtual Asset Service Providers (VASPs) under OJK (Otoritas Jasa Keuangan) or Bank Indonesia regulation, because they neither hold, transmit, nor interact with any digital asset at any point in the disbursement flow.
- The community hub’s only counterparty relationship is with their domestic bank and the Payment Gateway (Xendit), both of which are existing, regulated relationships predating ArPay entirely.

- The ArPay operator entity interacts with the Payment Gateway under standard business KYB (Know Your Business) terms, not crypto-specific licensing.

This makes ArPay the safest available mechanism for distributing sustainability grants and green incentive rewards to community-level environmental programs in Indonesia: organizations can receive borderless crypto-denominated funding without ever touching a wallet, acquiring a token, or assuming any regulatory classification as a crypto entity.

6. Formal Settlement Guarantee

We state the following guarantee informally. A rigorous formal proof is left for a subsequent version of this paper.

Theorem (Atomic Incentive Disbursement).

For any incentive grant transaction T accepted on-chain at slot s, one of the following two outcomes is guaranteed to occur within the timeout window W:

```
Let T = a confirmed ArPay incentive transaction at slot s.
Let W = the escrow timeout (120 seconds).
Let F = the event that the community hub receives IDR credit.
Let R = the event that the eco-sponsor receives USDC refund.

Guarantee:  P(F ∪ R | T confirmed) = 1
            P(F ∩ R | T confirmed) = 0

i.e., exactly one of {fiat disbursement, USDC refund} occurs.
```

The guarantee holds under the assumption that the Solana network achieves at least one non-partitioned epoch within W after slot s, and that the ArPay program's on-chain logic is correctly implemented. The program's source code is open-source and verifiable against the deployed program ID.

This guarantee is fundamental to ArPay's value proposition as green grant infrastructure: no sustainability program can be in a state where a sponsor's grant is deducted but a community hub receives nothing. Either the disbursement succeeds, or the eco-sponsor is made whole.

7. Implementation

7.1 Technology Stack

Component	Implementation	Rationale
On-Chain Program	Rust 1.75, Anchor 0.29	Type-safe, auditable Solana programs with built-in account validation
Client Application	Next.js 14 (PWA)	Mobile-first progressive web app; no native install required
Wallet Integration	@solana/wallet-adapter	Standard interface for Phantom, Solflare, and other non-custodial wallets
Grant Standard	Solana Pay Protocol	Open URI standard for reproducible, scannable incentive grant requests
Price Oracle	Pyth Network (on-chain)	First-party oracle; same-slot price updates; no round-trip latency
RPC Provider	Helius / QuickNode	Premium nodes required for WebSocket event delivery at < 200ms
Oracle Bridge	Python 3.11, asyncio	Async I/O for concurrent WSS listening and HTTP disbursement calls
Payment Gateway	Xendit Indonesia	Licensed by OJK; supports BI-FAST and all major Indonesian bank networks
Fiat Rail	BI-FAST (Bank Indonesia)	Domestic instant transfer network; < 2s settlement finality

7.2 Performance Characteristics

Under nominal conditions on Solana Mainnet-Beta with a premium RPC provider, the following latency distribution is observed across the incentive disbursement pipeline:

Stage	P50 Latency	P99 Latency
QR Scan → Wallet Prompt	< 500ms	< 1,200ms
Wallet Sign → Block Confirmed	400ms	1,200ms
Block Confirmed → WSS Event Received	80ms	400ms
WSS Event → Xendit API Response	300ms	900ms
Xendit Dispatch → BI-FAST Credit	800ms	2,000ms
Total (T0 → Community Hub Credit)	~2.5s	~6.0s

8. Conclusion

We have described ArPay, an Eco-Incentive Settlement Protocol that enables DePIN networks, sustainability sponsors, and environmental grant programs to disburse USDC-denominated rewards to QRIS-registered community hubs in real time — without requiring the community receiver to interact with any cryptocurrency infrastructure.

The protocol achieves this through a combination of Solana's programmable escrow model, a cryptographically-verified event bridge, and a licensed fiat disbursement layer. The key properties of the system are: non-custodial fund management for the eco-sponsor, complete regulatory transparency for community receivers (who remain entirely outside the definition of Virtual Asset Service Providers under OJK/BI regulation), and a formal atomic incentive disbursement guarantee that ensures either the community hub receives fiat or the eco-sponsor receives a full refund. No intermediate state exists in which funds are permanently lost.

The present implementation targets the Indonesian QRIS market and the rapidly growing DePIN and sustainability grant ecosystem. The protocol architecture is agnostic to the specific QR payment standard and fiat disbursement API, and can be adapted to any market with equivalent infrastructure — enabling a globally consistent layer for borderless green incentive distribution to local environmental communities.

Future work includes a formal security proof, support for multi-hop bridging across fiat corridors, native integration into the HYDRA enterprise POS ecosystem, and a programmatic grant-streaming API for continuous per-epoch DePIN node reward disbursements.

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